École polytechnique fédérale de Lausanne

Semester project spring 2026

Bachelor in Mathematics

Minimax Goodness-of-fit testing in High-Dimensional models

Author:
Hamza Lasri

Supervisor:
Sven Wang

Contents

Chapter 1

Introduction

Chapter 2

Minimax optimal testing by classification for Gaussian models

2.1 Setting of the problem

Definition 1. *Let μ, ν be two probability measures on a space $\mathcal{X}$, we define the total variation between μ and ν as follows :*

$$TV(\mu,\nu) = \sup_{E \subset \mathcal{X} measurable} |P(E) - Q(E)|$$

If μ, ν have densities f, g respectively with respect to another measure λ, then one has the equality :

$$TV(\mu,\nu) = \int |p(x) - q(x)| \, d\lambda(x)$$

Now, suppose that one has two samples X, Y of size n from two distributions $\mathbb{P}_X$ and $\mathbb{P}_Y$ respectively on some space $\mathcal{X}$ and wishes to test the hypotheses :

$$\mathcal{H}_0 : \mathbb{P}_X = \mathbb{P}_Y \text{ versus } \mathcal{H}_1 : TV(\mathbb{P}_X, \mathbb{P}_Y) \geq \varepsilon$$

for some $\varepsilon > 0$, by using machine learning classifiers. Given such hypotheses $\mathcal{H}_0, \mathcal{H}_1$ seen as families of distributions on $\mathcal{X}$, we say that a decision rule $\psi : \mathcal{X} \to \{0,1\}$ tests the two hypothesis against each other with error at most $\delta \in \left\{0, \frac{1}{2}\right\}$ if :

$$\max_{i=0,1} \max_{P \in \mathcal{H}_i} \mathbb{P}_{S\sim P}\Big(\psi(S) \neq i\Big) \leq \delta$$

The idea of using classifiers, firstly described by Friedman [2004], consists of training a binary classifier $\mathcal{C} : \mathcal{X} \to \{0,1\}$ on the labeled data $\cup_{i=1}^{n}\left\{(X_i, 0), (Y_i, 0)\right\}$ and compare the samples $\mathcal{C}(X_1), \ldots, \mathcal{C}(X_n)$ to $\mathcal{C}(Y_1), \ldots, \mathcal{C}(Y_n)$. **ADD MORE DETAILS**

We will follow the framework described by Gerber et al. [2023] as "Classifier-Accuracy testing" (CAT) for goodness-of-fit testing :

Suppose we have a sample X of size $2n$ from an unknown distribution $\mathbb{P}_X$ that we will split into 2 halves $X = X^{\text{tr}} \sqcup X^{\text{te}}$ and another sample Y of size $2n$ from a known distribution $\mathbb{P}_Y$ which corresponds to our null hypothesis, we will split this sample in the same manner as for the previous sample into $Y = Y^{\text{tr}} \sqcup Y^{\text{te}}$. We train a classifier $\mathcal{C} : \mathcal{X} \to \{0,1\}$ on the input $\mathcal{D}^{\text{tr}} :== \left\{X^{\text{te}}, Y^{\text{te}}\right\}$ that aims to assign 1 to $\mathbb{P}_X$ and 0 to $\mathbb{P}_Y$. In practice, it is easier to think of $\mathcal{C}$ in terms of the separating set $S := \mathcal{C}^{-1}(\{1\})$ which is a random set whose randomness come from the splitting of our samples and potentially an external seed.

Definition 2. *Given two datasets $\left\{A_i\right\}_{i=1}^{a}, \left\{B_j\right\}_{j=1}^{b}$, we define the classifier-accuracy stastistic s :*

$$T_S(A,B) := \frac{1}{a}\sum_{i=1}^{a} \mathbb{1}_{A_i \in S} - \frac{1}{b}\sum_{j=1}^{b} \mathbb{1}_{B_j \in S}$$

Finally, we threshold the output $|T_S|$ of the test data $\mathcal{D}^{\text{te}} := \left\{X^{\text{te}}, Y^{\text{te}}\right\}$

2.1.1 The Sobolev ellipsoid and Gaussian sequence model

In this project, we will restrict ourselves to a very specific class of distributions, the Gaussian sequence model with mean vector in the Sobolev ellipsoid that we will define below.

Definition 3. *Let $s > 0, C > 0$. We define the Sobolev ellipsoid $\mathcal{E}(s, C)$ as follows :*

$$\mathcal{E}(s, C) := \left\{ \theta \in \mathbb{R}^{\mathbb{N}} : \sum_{j=1}^{\infty} j^{2s} \theta_j^2 \leq C \right\}$$

For a sequence $\theta \in \mathcal{E}(s, C)$, we define $\mu_\theta = \bigotimes_{j=1}^{\infty} \mathcal{N}(\theta_j, 1)$. The set of Gaussian sequence model distributions is hence defined as :

$$\mathcal{P}_G(s, C_G) := \{\mu_\theta; \theta \in \mathcal{E}(s, C_G)\}$$

ADD MORE DETAILS ABOUT THE INTUITION BEHIND GAUSSIAN SEQUENCE MODEL

2.2 Preliminary results

The hard part of this kind of frameworks consists in finiding a good separating set $S \subset \mathcal{X}$, but we firstly need to define what "good" means.

Definition 4. *Let $S \subseteq \mathcal{X}$ be a measurable set, and μ, ν be two probability measures on $\mathcal{X}$. We define the separation and size of S respectively as follows :*

$$sep(S) = \mu(S) - \nu(S) \ \text{ and } \ \tau(S) = \min(\mu(S)\mu(S^c), \nu(S)\nu(S^c))$$

Lemma 1. *Gerber et al. [2023, Lemma 1]*
Consider the hypothesis testing problem $\mathcal{H}_0 : p = q$ versus any other alternative $\mathcal{H}_1$. Suppose that the learner has constructed a separating set S such that :

$$\forall \mu, \nu \in \mathcal{H}_1 : |sep(S)| = |\mu(S) - \nu(S)| \geq \underline{sep} \ \text{ and } \ \forall \mu, \nu \in \mathcal{H}_0 \cup \mathcal{H}_1 : \min(\mu(S)\mu(S^c), \nu(S)\nu(S^c)) \leq \bar{\tau}$$

Then, using only the knowledge of $\bar{\tau}$, the classifier-accuracy test with samples of size n from both p and q and an appropriate threshold achieves type-I and type-II errors at most δ provided that :

$$n \geq c \frac{\log(\frac{1}{\delta})}{\underline{sep}} \left(1 + \frac{\bar{\tau}}{\underline{sep}}\right)$$

for a large enough constant $c > 0$

Remark 1. *With Lemma 1, one can imagine that a separating can be referred to as "good" if $|sep(S)|$ is large under H_1, and $\tau(S)$ is small under both H_0 and $\mathcal{H}_1$ with probability $1 - \delta$*

In order to prove lemma 1, we will need a prior result on the Bernstein concentration of the classifier accuracy statistics.

Lemma 2. *Gerber et al. [2023, Lemma 8] Suppose $A_1, A_2, \ldots, A_n \overset{iid}{\sim} Ber(p)$ and $B_1, B_2, \ldots, B_m \overset{iid}{\sim} Ber(q)$. Let $\tau = \min(p(1-p), q(1-q))$, and let $\bar{A}, \bar{B}$ be their empirical means, respectively. There exists a universal constant $c > 0$ such that :*

$$\mathbb{P}\left(\left|\bar{A} - \bar{B}\right| \leq \frac{1}{2}|p - q| - \frac{1}{2}\sqrt{\frac{c\log(1/\delta)\tau}{\min(n, m)}} - \frac{1}{2}\frac{c\log(1/\delta)}{\min(n, m)} \right) \leq \delta,$$

$$\mathbb{P}\left(\left|\bar{A} - \bar{B}\right| \geq 2|p - q| + 2\sqrt{\frac{c\log(1/\delta)\tau}{\min(n, m)}} + 2\frac{c\log(1/\delta)}{\min(n, m)} \right) \leq \delta$$

Proof of lemma 1. Using n test samples from (X, Y), consider the following classifier-accuracy test: we accept $\mathcal{H}_0$ if

$$\left| \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}_{X_i \in S} - \mathbb{1}_{Y_i \in S} \right| \leq \sqrt{\frac{c\bar{\tau}\log(1/\delta)}{n}} + \frac{c\log(1/\delta)}{n}$$

and reject the null hypothesis otherwise. Above, $c > 0$ is a large absolut constant, and note that the threshold only relies on the knowledge of $\bar{\tau}, n$ and δ

To analyse type-I and type-II errors , first, assume that $\mathcal{H}_0$ holds. Since $\mathrm{sep}(S) = 0$ under the latter hypothesis, the second statement of Lemma 2 implies that we accept $\mathcal{H}_0$ with probability at least $1 - \frac{/delta}{2}$ if c is large enough. Moreover, if $\mathcal{H}_1$ holds, with probability at least $1 - \frac{\delta}{2}$, the first statement of Lemma 2 implies that :

$$\left| \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}_{X_i \in S} - \mathbb{1}_{Y_i \in S} \right| \geq |\underline{sep}| + \left(\sqrt{\frac{c\bar{\tau}\log(1/\delta)}{n}} + \frac{c\log(1/\delta)}{n} \right)$$

By the assumed lower bound $n \geq c\frac{\log(\frac{1}{\delta})}{\underline{sep}}\left(1 + \frac{\bar{\tau}}{\underline{sep}}\right)$, we will reject $\mathcal{H}_0$ as desired. $\square$

Chapter 3

Optimal transport based methods in Goodness-of-Fit problems

Chapter 4

Appendix

Chapter 5

Conclusion

Bibliography

Jerome Friedman. On multivariate goodness–of–fit and two–sample testing. 03 2004.

Patrik R. Gerber, Yanjun Han, and Yury Polyanskiy. Minimax optimal testing by classification. In Gergely Neu and Lorenzo Rosasco, editors, *Proceedings of Thirty Sixth Conference on Learning Theory*, volume 195 of *Proceedings of Machine Learning Research*, pages 5395–5432. PMLR, 12–15 Jul 2023. URL `https://proceedings.mlr.press/v195/gerber23a.html`.